

National Coverage Determination (NCD)

Blood Brain Barrier Osmotic Disruption for Treatment of Brain Tumors

110.20

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Tracking Information

Publication Number

100-3

Manual Section Number

110.20

Manual Section Title

Blood Brain Barrier Osmotic Disruption for Treatment of Brain Tumors

Version Number

1

Effective Date of this Version

03/20/2007

Implementation Date

05/07/2007

Description Information

Benefit Category

Incident to a physician's professional Service

Inpatient Hospital Services

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

A. General

The blood brain barrier (BBB) of the central nervous system is characterized by tight junctions between vascular endothelial cells, which prevent or impede various naturally occurring and synthetic substances (including anti-cancer drugs) from entering brain tissue. The BBB may be partly responsible for the poor efficacy of chemotherapy for malignant primary or metastatic brain tumors.

The BBBD is the disruption of the tight junctions between the endothelial cells that line the capillaries in the brain accomplished by osmotic disruption, bradykinin or irradiation. Theoretically, disruption of the BBB may, in the treatment of brain tumors, increase the concentration of chemotherapy drugs delivered to the tumor and may prolong the drug-tumor contact time.

Osmotic disruption of the BBB is the most common technique used. Chemotherapeutic agents are given in conjunction with barrier disruption. The BBBD process includes all items and services necessary to perform the procedure, including hospitalization, monitoring, and repeated imaging procedures.

Indications and Limitations of Coverage

B. Nationally Covered Indications

N/A

C. Nationally Non-Covered Indications

Effective for services performed on and after March 20, 2007, the Centers for Medicare & Medicaid Services determines that the use of osmotic BBBD is not reasonable and necessary when it is used as part of a treatment regimen for brain tumors.

D. Other

This NCD does not alter in any manner the coverage of anti-cancer chemotherapy.

(This NCD last reviewed March 2007.)

Transmittal Information

Transmittal Number

67

Coverage Transmittal Link

<https://www.cms.gov/Regulations-and-Guidance/Guidance/Transmittals/Downloads/R67NCD.pdf> 

Revision History

04/2007 - Effective for claims with dates of service on and after March 20, 2007, CMS determines that the use of osmotic blood brain barrier disruption is not reasonable and necessary when it is used as part of a treatment regimen for brain tumors. This NCD does not alter in any manner the coverage of anti-cancer chemotherapy. Effective date: 03/20/2007. Implementation date: 05/07/2007 ([TN67](#))

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for Blood Brain Barrier Disruption \(BBBD\) Chemotherapy \(CAG-00333N\)](#)

Additional Information

Other Versions

Title	Version	Effective Between	
Blood Brain Barrier Osmotic Disruption for Treatment of Brain Tumors	1	03/20/2007 - N/A	You are here